ECE 3113 - Energy Conversion

Homework 1

(Due on Friday 09/08/2023)

1. With the appropriate trigonometric identities and some math derivations, find the value of:

$$sin(\omega t) + sin(\omega t - 2\pi/3) + sin(\omega t + 2\pi/3)$$

Explain the engineering intuition behind this mathematic derivation.

2. Derive the instantaneous power for following instantaneous current and voltage.

$$v(t) = 140\, cos(120\pi t + 25^{\circ})$$
$$i(t) = 14\, cos(120\pi t - 78^{\circ})$$

3. Find the current I_L in the following scenarios:

1) Z=500 Ohm and is at location a)
2) Z=j500 Ohm and is at location b)
3) Z=-j500 Ohm and is at location c)